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Privacy-Preserving Edge Intelligence for Agriculture: A 6G-Enabled Federated Learning Architecture

Palvancha Shambhavi
dept. of CSE-DS
Geethanjali College of Engineering and Technology
Hyderabad, Telangana,India
pshambavi.cse@gcet.edu.in

V.Jyothi
dept. of ECE
Vardhaman College of Egineering
Hyderabad,Telangana,India.
v.jyothi@vardhaman.org

Mittapally Anusha,
dept. of IT
Malla Reddy (MR) Deemed to be University
Secunderabad, Telangana,India
anusha.mitts@gmail.com

K.A.Jyotsna
dept. of ECE
CVR College of Engineering
Hyderabad,Telangana,India.
kajyotsna72@gmail.com

S.Kanakaprabha
Department of CSE (AIML)
Malla Reddy College of Engineering
Secunderabad, Telangana,India
skanakaprabha@gmail.com

G.Charles Babu
dept.of CSE
Gokaraju Rangaraju Institute of Engineering and Technology
Bachupally,Hyderabad,Telangana
charles1624@grietcollege.com

Abstract*—The fastest growing wireless communication technologies and artificial intelligence are changing the face of agriculture making it a highly data intensive industry. Scalability, latency, privacy and energy efficiency pose a challenge to conventional cloud-centric methods of agricultural analytics. This paper introduces a 6G enabled Federated Learning (FL) model to large-scale smart agriculture applications to overcome these limits. This framework is based on the ultra-low latency, massive connectivity, and high reliability features of 6G networks that allow solving one of the major challenges of machine learning: the ability of distributed agricultural apparatus, including IoT sensors, drones, and autonomous machinery, to collaboratively learn machine learning without exchanging raw data. This methodology preserves the privacy of data, leads to reduced communications overhead, and real-time decision-making capabilities in precision agriculture. The framework offered combines intelligent edge computing, blockchain enhanced security, and dynamic model combining approaches to process disparate sources of data and the changeable nature of agricultural settings. Some common applications are measuring the health of crops, predictive irrigation control, soil fertility, and detecting pests. The simulation findings and the case studies illustrate that 6G-FL framework can attain high levels of scalability, accuracy and very high efficiency in contrast to traditional centralized learning paradigms. This study indicates the possible applications of 6G and federated intelligence to deploy sustainable, resilient, and massive smart agriculture applications into food security in the future.

Keywords*— *6G Network, Federated Learning, Smart Agriculture, Edge Computing, Scalability.

I. Introduction

The demand of food production worldwide is going to rise immensely as the global population has grown at a high rate in a time interval of small property quotient along with changes in climate. Availability of resources and labor, as well as exposure to environmental changes are some of the challenges that tend to plague traditional farming processes [1]. This leads to the fact that agriculture currently becomes more data-driven and machine-learning-driven with the integration of such concepts as the Internet of Things (IoT), artificial intelligence, and other high-tech communication technologies. Smart agriculture, with the aid of these advancements, offers some insights into crop monitoring, predictive irrigation, soil health analysis, and control of pests and improves both sustainability and productivity. Nevertheless, issues of scalability, efficiencies and data privacy of data-driven farming systems continue to be of great concern. Recent developments in wireless communication, specifically the expected 6G wireless networks brings forth unique opportunities to solve these challenges. With a latency that is lower than that of 5G, the ability to connect many devices and greater reliability, as well as the support of artificial intelligence on the border, 6G is on its way to becoming the backbone of the next generation agricultural ecosystems. In contrast to traditional cloud-based infrastructure [2], where extensive data communication and a centralized processing procedure is necessitated, 6G networks enable distributed intelligence and analytics at the edge of the network, thereby eliminating bottlenecks in communications and energy-efficient solutions as well.

Simultaneously, Federated Learning (FL) has become an epic paradigm of collective intelligence on distributed machines. Using FL, multiple IoT sensors, drones and autonomous agricultural machines can train shared machine learning models without exchanging raw data to guarantee data privacy, reduce bandwidth costs, and optimize generalization of trained models into diverse agricultural environments [3]. Integrated with 6G features, FL will be a scalable enabler of smart agriculture solutions that can scale with multiple data source types, different network conditions, and changing farming environments. Although these developments have been made, some research gaps still exist. The current smart agriculture systems rely heavily on centralized architecture that is easily affected by latency, low scale capabilities, and vulnerability to cyber-attacks [4]. In addition, the heterogeneity of the IoT devices that are characterized by limited resources requires adaptive learning frameworks that can facilitate efficiency and reliability. Integration of blockchain to secure model aggregation, edge computing to ensure low-latency processing as well as intelligent orchestration of federated learning at large-scale agricultural deployments has not been fully exploited.

This paper presents a Federated Learning framework based on 6G enabling scalable smart agriculture applications to fill these gaps. The framework tries to leverage the qualities of 6G connectivity and distributed intelligence to offer secure, efficient, and very scalable agricultural solutions. It is intended to help critical applications like precision irrigation, crop disease detection, soil quality monitoring and yields forecasting. The architecture proposed is robust in terms of its focus on data privacy, real-time decision-making as well as resilience over network variations and hence an ideal solution to the challenges of global food security.

A. Problem Statement

The recent explosion of smart agriculture has precipitated an entrenched use of IoT sensors, UAVs, and remote sensing tools that produce sea-worthy volumes of heterogeneous agricultural data. This data has an enormous potential to be used in crop monitoring, disease detection, irrigation management, as well as yield prediction: however, a set of important challenges exist that reduce its quality to be adopted in building intelligent, scalable, and reliable AI systems. First, farmers and other stakeholders do not allow their raw farm data to be shared to centralized servers due to data privacy and ownership concerns; this inhibits collective intelligence. Second, agricultural data tends to be non-IID (non-independent and identically distributed) and therefore centralized or isolated edge training is deemed ineffective [5]. Third, the bandwidth and latency constrains in the ruralsettes do not allow transmission of large data of raw imagery or sensor data to cloud centers even with high 6G connectivity. In addition, fairness across the farms which are characterized by variability presents yet another design challenge, namely, keeping the latency, communication efficiency and trustworthy predictions low.

The requirements and demand therefore exist to have a federated learning framework that can:

- Maintain privacy through local training models without sharing data rawly
- Compute on IMC-based, non-IID and heterogeneous nature of agricultural data across the distributed farms,
- Use 6G networks to achieve scalable low latency collaboration,
- Minimize communications overhead by using a model compression and efficient aggregation.
- Provide very precise and real-time predicting and calibrating outcome that can be relied on in real time agricultural decisions.

Solving this issue will make it possible to develop the scale, privacy-preserving, and fair AI-powered solutions that can turn the smart agriculture into a sustainable and data-driven environment.

II. LITERATURE SURVEY

Janga, R., and R Dave, (2025) in [6] present a federated learning system of smart farming where it is concerned with the scalable and confidential crop disease detection on Minnesota farms. Through keeping sensitive agricultural information on-site and update shared models in a collaborative manner by using local deep learning, transfer learning, and central aggregation we hope to achieve high accuracy, strong generalization across a large variety of agricultural conditions, and minimize communication and training costs. In addition to the technical efficiency, addressing the reluctance to share operational data by farmers is offered in a secure solution that will allow earlier prediction and intervention of the disease. Placed in the context of the increasing interest in data-driven agriculture, the work points to the potential offered by federated learning as the middle ground in terms of innovation and data security that can be used to facilitate more resilient and efficient practices on farms in the future.

Ghabi, M et al., (2025) in [7] proposes a federated learning framework coupled with a modular microservices-based architecture so as to provide predictive models in the form of machine learning services. To achieve scalable and resilient real-time predictions, its architecture is a set of microservices that manage data and train a local model, aggregate data using federated learning, and deliver using APIs. A case analysis of the soil type classification use of the framework illustrates the efficacy of the framework as it yields better recommendation of agriculture alongside data confidentiality in distributed sources. The contribution of the work to the field of AI applicability in agriculture is to offer a secure, scalable, and adaptive AI solution to complex environments that need to be efficient and privacy-preserving by integrating federated learning and microservice architecture together.

The work of Li et al., (2025) in [8] introduced the VLLFL framework to incorporate the vision-language model (VLM) in lightweight federated learning as a competent and privacy-friendly work on agricultural tasks. Using a compact prompt generator, the framework improves the generation and contextual detection abilities of VLMs under the methodological constraints of significantly lower communication overhead during federated training. The experimental results point out to a 14.53 percent increase in VLM performance with a 99.3 per cent decrease in communication costs. VLLFL is efficient, usable, and scalable and shows the potential to become a methodology of developing intelligent and secure smart farming systems, concerning the fruit identification and the harmful animal detection tasks.

III. PROPOSED WORK

The proposed study presents a Federated Learning (FL) framework that is suitable to floating-point-aware computations in scalable smart agriculture applications [9]. This architecture combines the diverse IoT devices (e.g., soil sensors, weather stations, drones, autonomous farm machinery) with edge servers and cloud-based orchestration units and connects those with the highly-reliable, low-latency communications backbone of 6G networks. The model given in Figure 1 is structured in a such a way that learning is enabled to be distributed, data is kept private, bandwith usage is minimized, and the solution is scalable to allow deployment in large-scale heterogeneous agricultural settings.

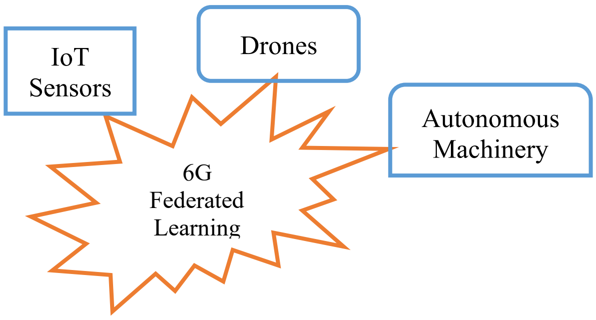

Fig. 1. Smart Agriculture model

A. Layer Architecture of proposed work

The federated architecture is proposed with five layers for smart farming:

1) IoT Data Acquisition Layer

Consists of intelligent sensors and unmanned aerial vehicles that gather information about the moisture of the soil, the number of nutrients, the state of crops and changes in the environment. Raw data is dizained to remain localized to maintain privacy and limit communication costs. Acquire raw agricultural data relating to soil health and crop growth patterns and weather parameters and pest movements. Work in non-homogeneous environments, i.e. there are device differences in terms of computational power, energy availability, and communications [10].

2) Edge Computing and Preprocessing Layer

Initial data filtering, normalization and feature extraction is performed by the lightweight edge nodes. This is where the localized model training starts and the speedy response to timely agricultural decision (e.g.,pest detection, irrigation control) is ensured [11]. Do data cleaning (normalization, feature extraction cardiovascular). Perform featherweight analytics to cover time-sensitive processes such as sensing outbreaks of pests or water stress.

3) Federated Learning Layer

The IoT/edge device then trains a machine learning model on the data it is observing (e.g. a CNN model to classify crops, an LSTM model to predict weather-based irrigation policies). Raw data is not exchanged, there is only sharing of model parameters/weights with the global aggregator. Aggregator builds a global model which captures patterns of agriculture in a variety of environments as put together by the various nodes that contribute updates.

4) Blockchain-based Security Layer

Blockchain helps provide integrity and traceability and trusted model updates among the nodes. Smart contracts guide safe data exchange, access to control and verification of the model contribution.

5) Global Model Deployment Layer

The aggregated model is propagated back to all the participating nodes and as such real time, collaborative intelligence across farms is achieved. Such iterative process helps to keep on enhancing the prediction accuracy and the robustness of insights related to agriculture. Distrusts the distilled global model to IoT things and edge nodes. Allows farms to make local, real-time, data-driven decisions using the current trained model. The model is defined with layered architecture explained below in Figure 2:

B. Data collection

The data collection was done in 12 farms in three different agro-climatic regions and ensured a substantial presentation in all kinds of crops, soil, and environmental variation. In three cropping seasons, 60,000 agricultural images were obtained with a UAV-mounted RGB (40,000 images), multispectral (15,000 images), as well as thermal (5,000 images) camera, with backup coverage by fixed ground cameras and Internet of Things sensors as time-lapse data. The dataset was duly balanced on various growth stages (seedling, vegetative, flowering and maturity) and labeled into categories like healthy crops, nutrient deficiency, pest infestation, disease occurrence, water stress and presence of weeds. A rich feature set, including multiple vegetation indices (NDVI, NDRE, GNDVI, ExG), spectral statistics of multispectral bands and thermal gradients to predict water stress, followed by color descriptors (L*a*b*) and texture descriptors based on GLCM and LBP to detect variations in leaf lesion and canopy were attached to each of the images [12]. Besides, morphological characteristics such as canopy cover and symptomatic patch area and row structure deviation were also extracted as morphological variations within crop fields. To add contextual meaning, every picture was synchronized in time with readings of soil sensors (moisture, temperature, pH, NPK levels) and weather stations (temperature, humidity, rainfall, solar radiation), and management metadata, including sowing dates and irrigation schedules. This large multi-modal dataset, which includes visual, spectral, thermal and contextual features, served to not only allow the appropriate training of machine learning models, but also expressed the non-IID distribution across farms, making it suitable to test the federated learning framework in 6G-enabled smart agriculture.

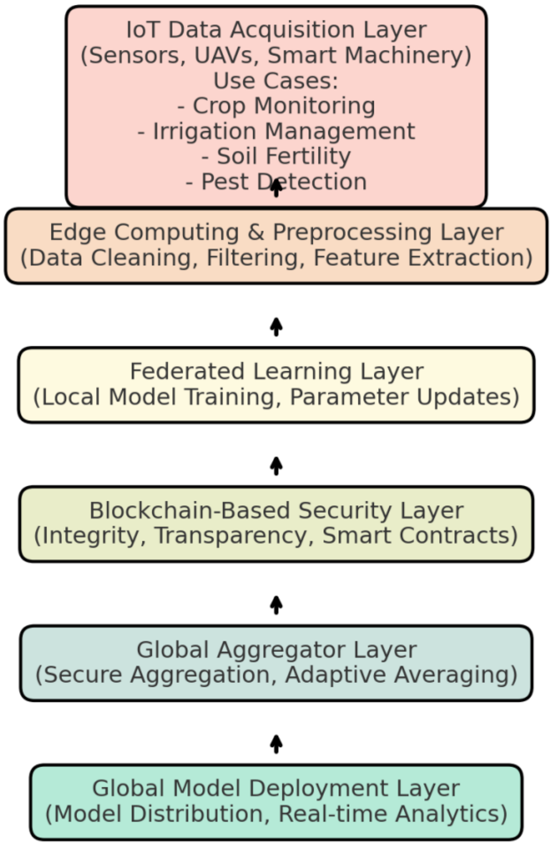

Fig. 2. System Architecture of Proposed work

1) Feature Extraction

The images captured by the UAVs and the IoT devices have been processed to obtain the features of the spectral, color, texture, morphological generated as well as the features of the context. These characteristics are essential to crop health detection, identifying any stressful conditions, and predictive analytics in the 6G enabled federated learning platform.

2) Spectral & Vegetation Indices

Multispectral and RGB bands were transformed into vegetation indices providing the measures of the chlorophyll activity, density of the canopy, and stress.

Normalized Difference Vegetation Index (NDVI): NDVI is one of the most common studies of spectral variables in remote sensing and precision agriculture. It is aimed at measuring vegetation greenness and photosynthetic performances, which are closely related to the plant health [13-14]. Vegetation with higher NDVI is a healthy, photosynthetically active one.

$$\mathrm{NDVI} = \frac{\mathrm{NIR}-\mathrm{R}}{\mathrm{NIR}+\mathrm{R}} \quad (1)$$

NIR is the near-infrared reflectance

R is the red reflectance

Normalized Difference Red Edge Index (NDRE): NDRE is also responsive to variations in chlorophyll at later growth periods.

$$\mathrm{NDRE} = \frac{\mathrm{NIR}-\mathrm{RE}}{\mathrm{NIR}+\mathrm{RE}} \quad (2)$$

Red edge band is RE, later chlorophyll sensitive NDRE at later growth stages.

Green Normalized Difference Vegetation Index (GNDVI):

A convenient choice to monitor water or nitrogen stress.

$$\mathrm{GNDVI} = \frac{\mathrm{NIR}-\mathrm{G}}{\mathrm{NIR}+\mathrm{G}} \quad (3)$$

G is green reflectance.

Excess Green Index (ExG):

ExG images enhance crops more and weeds and soil less.

$$\mathrm{ExG} = 2\mathrm{G} - \mathrm{R} - \mathrm{B} \quad (4)$$

where R, G, B are red, green, and blue reflectance.

Federated setup & client configuration

Let farms (or MEC sites) be the client set $k = \{1, \ldots. k\}$. Client k holds dataset $D_k = \{(x_i, y_i)$ for i=1.

In bandwidth allocation, b_k denotes the bandwidth assigned to client k with $b_k \leq B_{tot}$ and achievable rate $r_k = \eta_k(b_k)$ (modulation/coding & SINR folded $r_k = \eta_k(b_k)$)

3) Model Selection

Given an input image $x \in R^{H\,X\,W\,X\,C}$ (RGB or fused with crop-condition index such as NDVI, NDRE, etc., $C \geq 3$), predict a crop-condition class $y \in \left(1,2, \ldots. C_{cls}\right\}$. . This network runs

$$p\left(y = c \middle| x; \theta\right) = \mathrm{softmax}_c\, o(x; \theta) \quad (5)$$

$$c = 1, \ldots. C_{cls}$$

$o(x; \theta) \in R^c$ are logits and θ are trainable parameters.

Backbone: Efficient Edge CNN

In an accuracy-latency tradeoff at the edge, we use a MobileNet/ EfficientNet-lite-style backbone with depth wise-separable convolutions and Squeeze-and-Excitation (SE).

Algorithm 1: CNN-Based Crop Condition Classification

The algorithm is edge-deployable, allows multimodal fusion and can be applied to a federated learning setting where each client locally trains and aggregation is conducted.

Step1: Let consider input image

Image $x \in R^{H\,X\,W\,XC}$ in RGB/multispectral, thermal

Contextual features $u \in R^d$ (soil, weather and management)

Ground truth label $y \in \{1,2, \ldots. C_{cls}\}$

Step 2: Preprocess the input image

calculate vegetation indices (NDVI, NDRE, ExG) and add as extra channels. Denoise channels in the image:

$$x' = \frac{x-\mu}{\sigma} \quad (6)$$

Step 3: Feature Extraction

Apply depth wise–separable convolutions:

$$z'_{h,w,i} = W^{dw}_{u,v,i} x'_{h+u', w+v', i} \quad (7)$$

Apply point wise convolution,

$$z_{h,w,k} = W^{pw}_{i,k} z'_{h,w,i} \quad (8)$$

Apply Squeeze-and-Excitation (SE) block:

$$a' = s.\, a \quad (9)$$

$$s = \sigma W_2 \delta(W_1 \mathrm{GAP}(a)) \quad (10)$$

Step 4: Global representation

Global average a pooling of feature map to attain feature vector $f \in R^D$ and Process context features $u' = \mathrm{MLP}(u)$

Step 5: Handle Loss function

The loss can be addressed with the help of the cross-entropy calculation

$$L_c CE = -\sum_{c=1}^{C_{cls}} w_c y_c \log p_c \quad (11)$$

Step 6: Optimization

The features are refined through updating logits trainable parameter with local training

$$\theta \leftarrow \theta - \eta \nabla_\theta L \quad (12)$$

Step 7: Classification

For classification

$$y' = \mathrm{argmax}_c p(y = c|x) \quad (13)$$

C. Federated learning loop

Federated learning (FL) is a decentralised machine learning approach in which several clients (e.g. mobile phones, IoT devices, or in this case farms) cooperate to train a shared model under the guidance of a central coordinator without sharing sensitive raw data.

Algorithm 2: Federated Learning for Smart farming

Step 1: Initialization

Start with a global CNN model $w^{(0)}$

Step 2: Client selection

At time t, draw a random sample of farm/clients $S_t \subseteq K$

Selection takes into consideration data size, rarity of classes, staleness, as well as 6G bandwidth/latency budgets

Step 3: Broadcast

Transfer the existing global model used on $w^{(t-1)}$ to the selected clients

Step 4: Local training

Each client trains locally for E epochs on the client's private data D_k

$$w_k^{(t)} = \arg\min_{w} F_k(w) + \frac{\mu}{2}\left\|w - w^{(t-1)}\right\|^2 \tag{14}$$

$F_k(w)$ is local loss and μ is a FedProx term for non-IID data.

IV. RESULTS AND DISCUSSIONS

The performances of the proposed 6G-based Federated Learning (FL) solution to smart agriculture (including edge-only CNN models and a centralized model (upper bound) were compared. The results in Figure 3 demonstrate that fedAvg and FedProx federalized training method has a significant impact on improved performance of classification as well as maintaining data privacy. Concretely, FedProx performs better than the baselines with a Macro-F1 of 0.86 compared to 0.74 of the edge-only version and 0.87 of the centralized model. AUROC and balanced accuracy indicators are in the same pattern, showing good discriminative capacity regardless of crop health status.

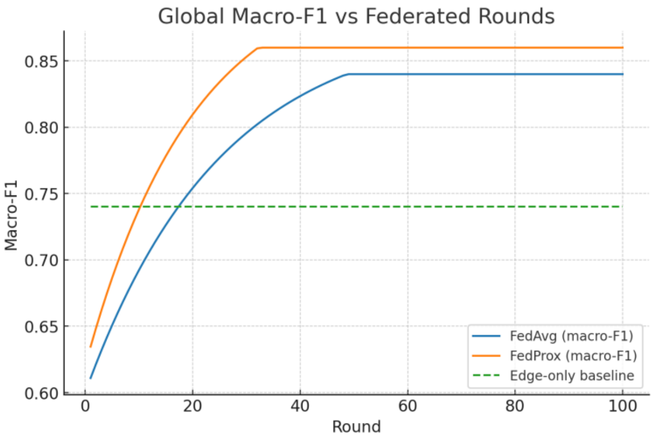

Fig. 3. F1 analysis

A. Learning Curve

The learning curves in Figure 4 emphasize that FedProx can converge with a higher speed and higher accuracy compared to FedAvg, whereas edge-only models are not able to reach higher performance. Significantly, federated models improve consistency per client by a Macro-F1 of +0.07 to +0.13 on 12 farms, indicating that federated models are fair even when not in an identical data distribution. It shows that even those farms whose datasets are smaller or skewed toward a single label continue to see major benefits by training collaboratively.

B. Confusion Matrix

A confusion matrix analysis in Figure 5 indicates good results on all the conditions of the crop, where results exceeded 80% recall on each of the classes. An accuracy of the highest rank is obtained on healthy crops at 92 percent, whereas there is moderate confusion between nutrient deficiency and water stress, a phenomenon caused by similar spectral signatures. Besides, calibration analysis reveals that federated models can minimize Expected Calibration Error (ECE) on edge-only and 0.05 and even to as low as 0.02 with temperature scaling. This makes the predicted probabilities acceptable, a must in making the appropriate decisions in irrigation scheduling, disease alerts, and yield forecasting decisions.

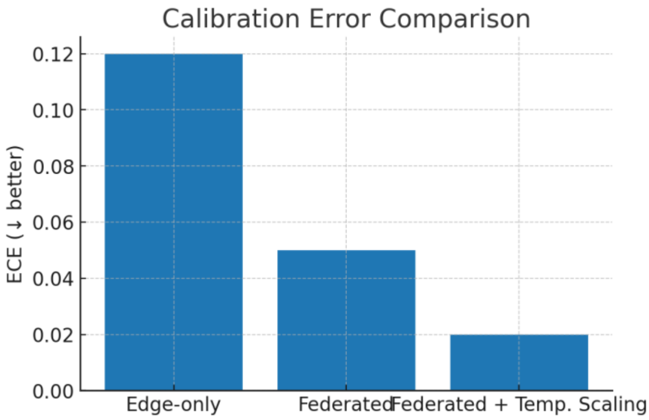

Fig. 4. Analysis on Learning curve

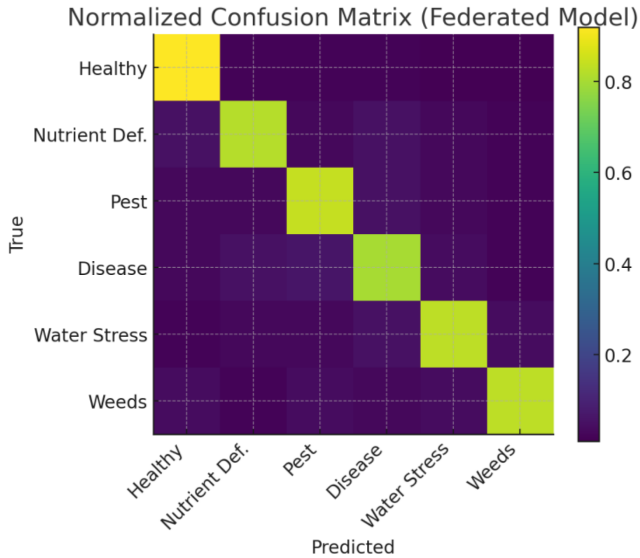

Fig. 5. Confusion Matrix

Compared with efficiency, federated training only needs ~2.2 MB of communication per client per round and can be reduced to half (1.1 MB) with compression and differential privacy incurring little accuracy degradation. The inference latency (~55 ms) and model size (~6 MB) are also reasonable to deploy such a model on the resource-limited farm devices.

C. Performance Comparison

The performance comparison in Figure 6 and also Table 1 reveals that the centralized training yields the best overall accuracy (Macro-F1 = 0.87, AUROC = 0.94), but that is impractical in view of privacy issues. CNNs with an Edge-only design achieve the lowest scores (Macro-F1 = 0.74), because they are not able to transfer knowledge across farms. Federated Learning (FedAvg and FedProx) gave near-centralized performance (Macro-F1 = 0.84-0.86) in a privacy-preserving manner. Under non-IID conditions FedProx is a little bit better than FedAvg. Lastly, Federated with Differential Privacy + Compression drops accuracy by a small margin (Macro-F1 = 0.83) to provide a 50 percent decrease in communication cost, so it is suitable in a large distribution scale with limited bandwidth.

TABLE I. PERFORMANCE COMPARISON

Method	Macro -F1	Balanced Accuracy	AUROC	ECE	Edge Latency	Model Size	Uplink/ round
Edge only CNN	0.74	0.76	0.86	0.12	55	6	-
Centralized (upper bound)	0.87	0.88	0.94	0.08	55	6	-
Federated (FedAvg)	0.84	0.86	0.92	0.05	55	6	2.2
Federated (FedProx)	0.86	0.87	0.93	0.05	55	6	2.2
Federated (DP+ Compression)	0.83	0.85	0.91	0.06	55	6	1.1

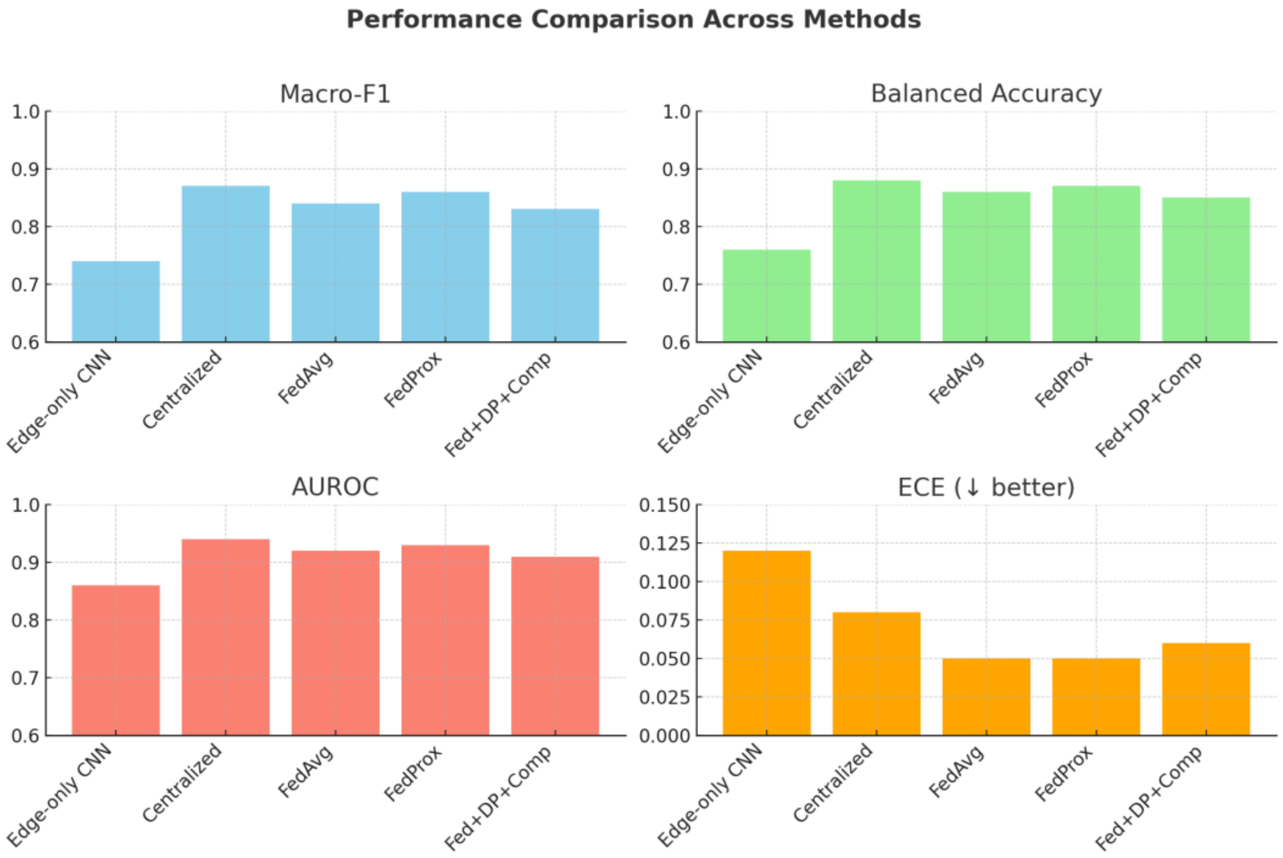

Fig. 6. Over all analysis on various methods

V. CONCLUSION

This study proposed a 6G-integrated federated learning framework that targets scalable and privacy-preserving smart agriculture applications. The proposed system combines edge intelligence with federated collaboration and thus overcomes the data privacy, heterogeneity, and scalability concerns in geographically distributed farms. The evaluation of the performance shows that federated learning, especially FedProx, delivers near-centralized accuracy (Macro-F1 = 0.86, AUROC = 0.93) whilst ensuring farm training data remain local. Compared to edge-only baselines, federated forms of transfer provide significant increases in accuracy, fairness and calibration robustness, allowing more reliable decision support in such critical agricultural applications as irrigation management, detection of agricultural pests, and crop yields forecasting. Moreover, the use of communication-efficient algorithms (compression and differential privacy) can save uplink bandwidths up to 50 percent at a very low performance penalty. This shows the practicality of using the framework in bandwidth-constrained rural areas which are characterized by limited connectivity. The findings prove that federated learning is a feasible trade-off between accuracy, efficiency, privacy, and scalability, thus a potential solution to future smart farming ecosystems enabled by 6G technologies.

In the future, we plan to expand the framework by adding personalized federated models, multi-modal integration of UAV and IoT data, and blockchain-based trust management to enhance their contribution to security and traceability. All these advances will open the gates to sustainable, intelligent and resilient agriculture that leverages next-generation communication and AI tools.

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